3.16

9.7.1.(3) 微元法 $dl = \sqrt{(dr)^2+(r\,d\theta)^2} = \sqrt{(\frac{dr}{d\theta})^2+r^2}\,d\theta$

$$= \sqrt{(-a\sin\theta)^2+a^2(1+\cos\theta)^2}\,d\theta = a\sqrt{2+2\cos\theta}\,d\theta$$

$$= a\cdot 2\left|\cos\frac{\theta}{2}\right|d\theta = 2a\left|\cos\frac{\theta}{2}\right|d\theta$$

$$l = \int_0^{2\pi} 2a\left|\cos\frac{\theta}{2}\right|d\theta = \int_0^{\pi} 2a\cos\frac{\theta}{2}\,d\theta - \int_{\pi}^{2\pi} 2a\cos\frac{\theta}{2}\,d\theta$$

$$= 2\int_0^{\pi} 2a\cos\frac{\theta}{2}\,d\theta = 4a\int_0^{\pi}\cos\frac{\theta}{2}\,d\theta = 8a\cdot\sin\frac{\theta}{2}\Big|_0^{\pi} = 8a$$

(4) 设 $y = at$ 有 $t\in[-\sqrt{2},\sqrt{2}]$. $x = \frac{a}{2}t^2$

$$dl = \sqrt{(dx)^2+(dy)^2} = a\sqrt{1+t^2}\,dt$$

$$l = \int dl = \int_{-\sqrt{2}}^{\sqrt{2}} a\sqrt{1+t^2}\,dt = a\int_{-\sqrt{2}}^{\sqrt{2}}\sqrt{1+t^2}\,dt$$

$$= a\int_{-\arctan\sqrt{2}}^{\arctan\sqrt{2}}\sqrt{1+\tan^2\theta}\,\frac{d\theta}{\cos^2\theta} = a\int_{-\arctan\sqrt{2}}^{\arctan\sqrt{2}}\frac{d\theta}{\cos^3\theta}$$

$$= a\int_{-\sqrt{\frac{2}{3}}}^{\sqrt{\frac{2}{3}}}\frac{d(\sin\theta)}{\cos^4\theta} = a\int_{-\sqrt{\frac{2}{3}}}^{\sqrt{\frac{2}{3}}}\frac{du}{(1-u^2)^2} = a\int_{-\frac{\sqrt{6}}{3}}^{\frac{\sqrt{6}}{3}}\frac{1}{4}\left(\frac{1}{1-u}+\frac{1}{1+u}+\frac{1}{(1-u)^2}+\frac{1}{(1+u)^2}\right)du$$

$$= \frac{1}{4}a\left(2\ln\frac{1+\frac{\sqrt{6}}{3}}{1-\frac{\sqrt{6}}{3}} + 2\left(\frac{1}{1-\frac{\sqrt{6}}{3}} - \frac{1}{1+\frac{\sqrt{6}}{3}}\right)\right) = \frac{a}{2}\left(\ln\frac{3+\sqrt{6}}{3-\sqrt{6}} + 2\sqrt{6}\right) = (\sqrt{6}+\ln(\sqrt{2}+\sqrt{3}))\,a$$

2.(6) $S = \left|\int_0^{2\pi} y\,x'\,dt\right| = \left|\int_0^{2\pi} a\sin^3 t\cdot 3a\cos^2 t\,(-\sin t)\,dt\right|$

$$= \left|\int_0^{2\pi} 3a^2\sin^4 t\cos^2 t\,dt\right| = 3a^2\left|\int_0^{2\pi}\frac{1+\cos 2t}{2}\left(\frac{1-\cos 2t}{2}\right)^2 dt\right|$$

$$= \frac{3}{8}a^2\left|\int_0^{2\pi}(1+\cos 2t)(1-\cos 2t)^2\,dt\right|$$

$$= \frac{3}{8}a^2\left|\int_0^{2\pi}(1-\cos 2t-\cos^2 2t+\cos^3 2t)\,dt\right|$$ 由对称性 $\int_0^{2\pi}\cos 2t\,dt = 0$, $\int_0^{2\pi}\cos^3 2t\,dt = 0$

$$\int_0^{2\pi}\cos^2 2t\,dt = \int_0^{2\pi}\frac{1+\cos 4t}{2}\,dt = \pi$$

$$\Rightarrow S = \frac{3}{8}a^2\left|2\pi-\pi\right| = \frac{3\pi a^2}{8}$$

3.21. $S=\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 2\pi(-a\cos t)\sqrt{x'^2+y'^2}\,dt$

$$=-2\pi a\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}\cos t\cdot a\sqrt{(1-\cos t)^2+\sin^2 t}\,dt=-2\pi a^2\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}\cos t\cdot\sqrt{2-2\cos t}\,dt$$

$$=-4\pi a^2\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}\cos t\cdot\sin\frac{t}{2}\,dt=-2\pi a^2\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}}\left(\sin\frac{3t}{2}-\sin\frac{t}{2}\right)dt$$

$$=-2\pi a^2\left(-\frac{2}{3}\cos\frac{3t}{2}\Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}}+2\cos\frac{t}{2}\Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}}\right)=-2\pi a^2\left(-\frac{2\sqrt{2}}{3}-2\sqrt{2}\right)=\frac{16\sqrt{2}\pi a^2}{3}$$

5.(3). 参数方程 $y=2at$, $x=2at^2$ 有 $t\in\left[-\sqrt{\frac{b}{2a}},\sqrt{\frac{b}{2a}}\right]$

$$V=\int\pi(b-x)^2dy=\int_{-\sqrt{\frac{b}{2a}}}^{\sqrt{\frac{b}{2a}}}\pi(b-2at^2)^2\cdot 2a\,dt=2\pi a\int_{-\sqrt{\frac{b}{2a}}}^{\sqrt{\frac{b}{2a}}}(b-2at^2)^2\,dt$$

$$=2\pi a\left(b^2\cdot 2\left(\frac{b}{2a}\right)^{\frac{1}{2}}-4ab\cdot\frac{1}{3}\cdot 2\cdot\left(\frac{b}{2a}\right)^{\frac{3}{2}}+4a^2\cdot\frac{1}{5}\cdot 2\cdot\left(\frac{b}{2a}\right)^{\frac{5}{2}}\right)$$

$$=2\pi a\left(2b^2-\frac{4}{3}ab\cdot\frac{b}{2a}+\frac{8}{5}a^2\cdot\frac{b^2}{4a^2}\right)\sqrt{\frac{b}{2a}}=\frac{16\sqrt{2}\pi a^{\frac{1}{2}}b^{\frac{5}{2}}}{15}$$

8.(2). 交点在 $(0,0)$ 和 (a,a). $y_1=\sqrt{ax}$, $y_2=\frac{x^2}{a}$ $x\in[0,a]$ 时有 $y_1\geq y_2$

$$\bar{x}=\frac{\int_0^a x(y_1-y_2)\,dx}{\int_0^a(y_1-y_2)\,dx}=\frac{\int_0^a x\left(\sqrt{ax}-\frac{x^2}{a}\right)dx}{\int_0^a\left(\sqrt{ax}-\frac{x^2}{a}\right)dx}=\frac{\sqrt{a}\cdot\frac{2}{5}a^{5/2}-\frac{1}{4a}\cdot a^4}{\sqrt{a}\cdot\frac{2}{3}a^{3/2}-\frac{1}{3a}a^3}=\frac{9}{20}a$$

由关于 $y=x$ 的对称性可知 $\bar{y}=\frac{9}{20}a$ 故质心 $\left(\frac{9}{20}a,\frac{9}{20}a\right)$

补1: 以 θ 为参数有 $x=r(\theta)\cos\theta$, $y=r(\theta)\sin\theta$

由 $r(\theta)\in C^1[\alpha,\beta]$ 有 $x'=r'\cos\theta-r\sin\theta$, $y=r'\sin\theta+r\cos\theta$ 有 $x,y,x',y'\in R[\alpha,\beta]$

再由题中公式有 $S=\frac{1}{2}\int_\alpha^\beta(xy'-yx')\,d\theta$

$$=\frac{1}{2}\int_\alpha^\beta\left(r\cos\theta(r'\sin\theta+r\cos\theta)-r\sin\theta(r'\cos\theta-r\sin\theta)\right)d\theta$$

$$=\frac{1}{2}\int_\alpha^\beta r^2\,d\theta \quad \text{故证}$$

3.18

10. 1. 2. $x\in[0,1)$ 时 $\frac{1}{\sqrt{1-x^4}}<\frac{1}{\sqrt{1-x^2}}$，故 $\int_0^1\frac{dx}{\sqrt{1-x^4}}<\int_0^1\frac{dx}{\sqrt{1-x^2}}=\int_0^{\frac{\pi}{2}}\frac{d(\sin\theta)}{\cos\theta}=\frac{\pi}{2}$

$\frac{1}{\sqrt{1-x^4}}=\frac{1}{\sqrt{1-x^2}}\cdot\frac{1}{\sqrt{1+x^2}}>\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{1-x^2}}$ 故 $\int_0^1\frac{dx}{\sqrt{1-x^4}}>\frac{\sqrt{2}}{2}\int_0^1\frac{dx}{\sqrt{1-x^2}}=\frac{\sqrt{2}}{4}\pi$

5. (1) $|\cos x^2|\geq\cos^2x^2$ 故只需证 $\int_0^{+\infty}\cos^2x^2dx$ 发散即可证 $\int_0^{+\infty}|\cos x^2|dx$ 发散

$$\int_0^{+\infty}\cos^2x^2dx=\int_0^{+\infty}\cos^2t\,d(\sqrt{t})=\int_0^{+\infty}\frac{\cos^2t}{2\sqrt{t}}dt=\int_0^{+\infty}\frac{1+\cos2t}{4\sqrt{t}}dt$$

$\int_0^{+\infty}\frac{1}{4\sqrt{t}}dt=\frac{1}{2}\sqrt{t}\Big|_0^{+\infty}$ 发散

$$\int_0^{+\infty}\frac{\cos2t}{4\sqrt{t}}dt=\frac{\sqrt{2}}{8}\int_0^{+\infty}\frac{\cos u}{\sqrt{u}}du=\frac{\sqrt{2}}{8}\left(\int_0^1\frac{\cos u}{\sqrt{u}}du+\int_1^{+\infty}\frac{\cos u}{\sqrt{u}}du\right)$$

由 $u\in(0,1]$ 时 $\frac{\cos u}{\sqrt{u}}>0$ 且 $\int_0^1\frac{\cos u}{\sqrt{u}}du<\int_0^1\frac{du}{\sqrt{u}}=2\sqrt{u}\Big|_0^1=2$ 收敛

$$\int_1^{+\infty}\frac{\cos u}{\sqrt{u}}du=\int_1^{+\infty}\frac{d(\sin u)}{\sqrt{u}}=\frac{\sin u}{\sqrt{u}}\Big|_1^{+\infty}+\int_1^{+\infty}\sin u\frac{du}{2u^{3/2}}=-\sin1+\int_1^{+\infty}\frac{\sin u}{2u^{3/2}}du$$

$B>A>1$ 时 $\left|\int_A^B\frac{\sin u}{2u^{3/2}}du\right|\leq\int_A^B\frac{1}{2u^{3/2}}du=\frac{1}{\sqrt{A}}-\frac{1}{\sqrt{B}}\to0\ (A,B\to+\infty)$

$\Rightarrow\int_1^{+\infty}\frac{\sin u}{2u^{3/2}}du$ 收敛 $\Rightarrow\int_1^{+\infty}\frac{\cos u}{\sqrt{u}}du$ 收敛

7. 由 Cauchy 收敛准则，$\forall\varepsilon>0$，$\exists N_1>a$ s.t $B_1>A_1>N_1$ 时 $\left|\int_{A_1}^{B_1}g\,dx\right|<\varepsilon$

$\exists N_2>a$ s.t $B_2>A_2>N_2$ 时 $\left|\int_{A_2}^{B_2}h\,dx\right|<\varepsilon$

故取 $N=\max\{N_1,N_2\}$ 当 $B>A>N$ 时 $\int_A^Bg\,dx\leq\int_A^Bf\,dx\leq\int_A^Bh\,dx$

$\Rightarrow\left|\int_A^Bf\,dx\right|\leq\max\left\{\left|\int_A^Bg\,dx\right|,\left|\int_A^Bh\,dx\right|\right\}<\varepsilon$

由 Cauchy 收敛准则，f 在 $[a,+\infty)$ 上无穷限积分收敛

8. 若 $\exists x_0\in[a,+\infty)$ 使 $f(x_0)<0$ 则 $x\geq x_0$ 时恒有 $f(x)\leq f(x_0)$

则 $\left|\int_{x_0}^{+\infty}f(x)dx\right|>|f(x_0)|\cdot\int_{x_0}^{+\infty}dx\to+\infty$ 发散 矛盾

故始终有 $f(x)\geq0$ 单调有界存在极限 $\lim\limits_{x\to+\infty}f(x)=l\geq0$

有 $\forall x\in[a,+\infty)$ $f(x)\geq l$ 故 $\int_a^{+\infty} f(x)\,dx \geq l\int_a^{+\infty} 1\,dx$

由 $\int_a^{+\infty} f(x)\,dx$ 收敛可知 $l=0$ 即 $\lim\limits_{x\to+\infty} f(x)=0$ (*)

由 Cauchy 收敛准则，$\forall \varepsilon>0$ $\exists N_1\geq a$ 使 $B>A>N_1$ 时 $0<\int_A^B f(x)\,dx<\frac{\varepsilon}{2}$

$\Rightarrow f(B)(B-A)\leq \int_A^B f(x)\,dx<\frac{\varepsilon}{2}$

固定 A 由(*) $\exists N_2>a$ 使 $B>N_2$ 时 $f(B)<\frac{\varepsilon}{2A}$ 故 $f(B)\cdot A<\frac{\varepsilon}{2}$

$\Rightarrow f(B)\cdot B<\varepsilon \Rightarrow \lim\limits_{x\to+\infty} xf(x)=0$

9. $\forall\varepsilon>0$, $\exists\delta$ s.t $\forall |x_1-x_2|<\delta$ 均有 $|f(x_1)-f(x_2)|<\frac{\varepsilon}{2}$

由 Cauchy 收敛准则，$\exists N>0$ s.t $\forall A\geq N$ 均有 $\left|\int_A^{A+\delta} f\,dx\right|<\frac{\varepsilon\delta}{2}$

故由积分第一中值定理有 $|f(\xi)|\delta<\frac{\varepsilon\delta}{2}\Rightarrow |f(\xi)|<\frac{\varepsilon}{2}$ $(\xi\in(A,A+\delta))$

$\forall x\in(A,A+\delta)$，$|f(x)|\leq|f(\xi)|+|f(x)-f(\xi)|<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon$

由 A 的任意性可知 $\forall x>N$ 均有 $|f(x)|<\varepsilon\Rightarrow \lim\limits_{x\to+\infty} f(x)=0$

10.2.1 (5) 记 $F(x)=\int_1^x \frac{t\arctan t}{1+t^3}\,dt$. 由于 $\frac{t\arctan t}{1+t^3}>0$, $t\in[1,+\infty)$

故 $F(x)$ 为单调增函数，又 $\frac{t\arctan t}{1+t^3}<\frac{\pi}{2}\frac{t}{1+t^3}$

故只用证 $\int_1^{+\infty}\frac{x}{1+x^3}\,dx$ 收敛即可证 $F(x)$ 有上界进而有极限 $\Rightarrow$ 收敛.

$$\int_1^{+\infty}\frac{x}{1+x^3}\,dx=\int_1^{+\infty}\frac{1}{3}\left(\frac{x+1}{x^2-x+1}-\frac{1}{x+1}\right)dx=\frac{1}{3}\int_1^{+\infty}\left(\frac{(x-\frac{1}{2})+\frac{3}{2}}{(x-\frac{1}{2})^2+\frac{3}{4}}-\frac{1}{x+1}\right)dx$$

$$=\frac{1}{3}\left(\ln\frac{\sqrt{x^2-x+1}}{x+1}+\sqrt{3}\arctan\left(\frac{x-\frac{1}{2}}{\frac{\sqrt{3}}{2}}\right)\right)\Bigg|_1^{+\infty}$$

$$=\frac{1}{3}\left(\ln\sqrt{1-\frac{3}{x+1}+\frac{3}{(x+1)^2}}+\sqrt{3}\arctan\left(\frac{2x-1}{\sqrt{3}}\right)\right)\Bigg|_1^{+\infty}=\frac{1}{3}\ln 2+\frac{\sqrt{3}\pi}{9}$$ 收敛.

故 $\int_1^{+\infty}\frac{x\arctan x}{1+x^3}\,dx$ 收敛

(9) $\int_1^{+\infty}\frac{1}{x}\arctan\left(\frac{1}{x}\right)dx=\int_1^0 t\arctan t\,d\left(\frac{1}{t}\right)=\int_0^1\frac{\arctan t}{t}\,dt$

由 $t\in[0,1]$ 时 $\tan t>t$ 故 $t>\arctan t$，又 $f(A)=\int_A^1\frac{\arctan t}{t}\,dt$ 关于 $A\uparrow$ $(A\in[0,1])$

有 $\forall A\in[0,1)$，$f(A)<\int_{1-A}^1 dt<A<1$ 故有上界 $\Rightarrow$ 有极限 $\Rightarrow \int_0^1\frac{\arctan t}{t}\,dt$ 收敛

补1:

即证明 $\lim\limits_{A\to 0^+}\left(\int_{-1}^{-A} f(x)\,dx+\int_A^1 f(x)\,dx\right)=\lim\limits_{A\to 0^+}\int_A^1 (f(x)+f(-x))\,dx$ 收敛

相当于 $\int_0^1 (f(x)+f(-x))\,dx=\int_0^1\left(\frac{1}{\sin x}-\frac{1}{x}\right)dx$ 收敛

由Taylor展开 $\sin x = x-\frac{1}{6}x^3+\frac{1}{120}\cos\xi\, x^5 \quad \xi\in(0,x)$

可知 $x\in(0,1)$ 时 $x-\frac{1}{6}x^3<\sin x<x \Rightarrow 0<\frac{1}{\sin x}-\frac{1}{x}<\frac{x}{6-x^2}=\frac{1}{2}\left(\frac{1}{\sqrt{6}-x}-\frac{1}{\sqrt{6}+x}\right)$

记 $F(A)=\int_{1-A}^1\left(\frac{1}{\sin x}-\frac{1}{x}\right)dx$，$A\in[0,1)$ 有 $F(A)\uparrow$ 只需证 $F(A)$ 有上界即可证收敛

$F(A)<\int_{1-A}^1\frac{1}{2}\left(\frac{1}{\sqrt{6}-x}-\frac{1}{\sqrt{6}+x}\right)dx<\int_0^1\frac{1}{2}\left(\frac{1}{\sqrt{6}-x}-\frac{1}{\sqrt{6}+x}\right)dx=-\ln\sqrt{6-x^2}\Big|_0^1=\ln\sqrt{\frac{6}{5}}$ 有上界

$\Rightarrow \lim\limits_{A\to 0^+}F(A)$ 存在 $\Rightarrow \int_0^1\left(\frac{1}{\sin x}-\frac{1}{x}\right)dx$ 收敛